



East West University
Department of Computer Science and Engineering
Course Outline
Fall 2024 Semester

Course: CSE103 Structured Programming

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	1.5	4.5
Contact Hours	Will be advised in class	Will be advised in class	Will be advised in class

Prerequisite

None

Instructor Information

Instructor: Md Ashraful Haider Chowdhury, Lecturer, Dept of CSE.
Office: Room #1003 (AB3 building)
Tel. No.: To be announced later
E-mail: ashraful.chowdhury@ewubd.edu
TA: To be announced later

Office Hour

Day	Time
Sunday	12:15 PM – 2:55 PM
Monday	10:50 AM – 1:30 PM
Tuesday	12:15 PM – 2:55 PM
Thursday	09:25 AM – 10:40 AM 12:15 PM – 1:30 PM

Course Objective

The purpose of this course is to introduce the students to computer programming using structured language. The students will be able to enhance their analyzing and problem-solving skills and use the same for writing programs using C language. Knowledge of this course will be needed as prerequisite knowledge for future courses such as CSE106 Discrete Mathematics, CSE110 Object Oriented Programming, CSE207 Data Structures, CSE246 Algorithms, CSE302 Database Systems, CSE366 Artificial Intelligence, CSE405 Computer Networks and many others.

Knowledge Profile

K2: Conceptually-based mathematics, numerical analysis, statistics, and formal aspects of computer and information science

Learning Domains

Cognitive - C2: Understanding, C3: Applying
Psychomotor - P2: Manipulation, P3: Precision
Affective - A2: Responding

Program Outcomes (POs)

PO1: Engineering Knowledge

Complex Engineering Problem Solution

EP1: Depth of knowledge required
EP2: Range of conflicting requirements

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	Complex Engineering Problem Solving/ Engineering Activities
CO1	Understand and apply the fundamentals of programming, basics of elementary programming, and different control statements in the target language.	PO1	C2, C3	K2	EP1
CO2	Understand and apply the different types of arrays and functions for implementing structured programs.	PO1	C2, C3	K2	EP1, EP2
CO3	Understand different data structures like pointers, structures, unions, user defined data types, and dynamic memory for implementing structured programs.	PO1	C3	K2	EP1, EP2
CO4	Demonstrate skills to choose appropriate language constructs and data structures to design, build and test realistic, complex application.	PO1	C2, C3 P2, P3 A2	K2	EP1, EP2

Course Topics, Teaching-Learning Method, and Assessment Scheme

Course Topic	Teaching- Learning Method	CO	Mark of Cognitive Learning Levels		CO Mark	Exam (Mark)
			C2	C3		
Introduction to computers and programming languages, data representation in computer, flowchart construction for problem solving	Lectures, Class Discussions, Discussions Outside Class	CO1	2.5	2.5	5.0	Midterm Exam (30)
Introduction to C Programming (input, output, variables, data types, operators, expressions, assignments)	Do	CO1	2.5	2.5	5.0	
Conditional control statements (if, if-else, nested if-else, switch)	Do	CO1	2.5	2.5	5.0	
Loop statement (while, for and do...while), break and continue statements	Do	CO1	3.0	3.0	5.0	
Introduction to functions (function definitions, function prototypes and argument, header files). Solving complex problems in modular fashion using user defined function	Do	CO2		3.0	3.0	
Introduction to arrays (arrays, declaring arrays, manipulating arrays)	Do	CO2	3.0	4.0	7.0	
Nested loop statement	Do	CO1	2.5	3.5	6.0	Final (30)
Multidimensional array	Do	CO2	3.5	3.5	7.0	
Characters and strings (various types of string manipulation)	Do	CO2	2.5	3.5	5.0	
Introduction to recursive definition and solving problem using recursive function	Do	CO2		2.5	2.5	
Pointers (pointer variable declarations, pointer operators, passing arguments to functions by reference with pointers, pointer expressions and pointer arithmetic, arrays of pointers, and function pointers)	Do	CO3		2.5	2.5	

Course Topic	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels		CO Mark	Exam (Mark)
			C2	C3		
Structures (structure definitions and initialization, accessing structure members, structure with function and pointer)	Do	CO3		2.5	2.5	
File management (files and streams, creating a file, reading data from file, writing data to file, and updating files)	Do	CO3		2.5	2.5	
Dynamic memory allocation and linked lists	Do	CO3		2.0	2.0	

Laboratory Experiments and Assessment Scheme

Experiment	Teaching-Learning Method	CO	Mark of Cognitive Learning Levels		Mark of Psychomotor Learning Levels		Mark of Affective Learning Levels	CO Mark
			C2	C3	P2	P3	A2	
Problem solving using arithmetic operators and conditional control statements	Discussion, Report Writing, Coding and Running Program	CO4	0.5				0.5	1
Problem solving using loops	Do	CO4	0.5		0.5			1
Problem solving requiring array manipulation	Do	CO4			0.5		0.5	1
Problem solving requiring nested loop	Do	CO4		0.5		0.5		1
Problem solving requiring multi-dimensional array	Do	CO4			0.5	0.5		1

Problem solving requiring user defined function and string manipulation	Do	CO4				0.5	0.5	1
Problem solving involving file input/output	Do	CO4		0.5	0.5	0.5	0.5	2
Problem solving requiring user defined data types	Do	CO4		0.5	0.5	0.5	0.5	2
Lab Final Exam	Individual Lab Exam	CO4		2.5	2.5	2.5	2.5	10
Total			1.0	4.0	5.0	5.0	5.0	20

Mini Project

Mini Project	Teaching-Learning Method	CO	Mark of Cognitive Learning Level	Mark of Psychomotor Learning Levels		Mark of Affective Learning Level	CO Mark
			C3	P2	P3	A2	
Mini Project including Report and Presentation	Group-based, moderately complex C program for practical application with report writing and presentation	CO4	7	1	1	1	10

Overall Assessment Scheme

Assessment Area	CO				Other	PO Marks
	CO1	CO2	CO3	CO4		PO1
Class Test/Quiz					10	10
Midterm	15	15				30
Final Exam		15	15			30
Lab Performance				10		10
Lab Final				10		10
Mini Project				10		10
Total	15	30	15	30	10	100

Teaching Materials/Equipment

Text book:

1. Teach Yourself C by Herbert Schildt
2. *Programming in ANSI C* by E. Balagurusamy

Lab Manual:

Lab manual will be provided in google classroom.

Project Description:

Project description will be provided google classroom.

Equipment/Software:

Any C/C++ IDE: CodeBlocks, Visual Studio Code, Visual C++, and/or Dev-C++

Exam Dates

Section	Midterm	Final
19/20/28	Will be notified in the class	See EWU calendar

*** Exam schedule may change as per the schedule provided by the university.

Grading System

Marks (%)	Letter Grade	Grade Point	Marks (%)	Letter Grade	Grade Point
80-100	A+	4.00	55-59	B-	2.75
75-79	A-	3.75	50-54	C+	2.50
70-74	B+	3.50	45-49	C	2.25
65-69	B	3.25	40-44	D	2.00
60-64	B	3.00	0-39	F	0.00

Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offence under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student **MUST** have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter into the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student misses any exam, the

student MUST get approval of makeup exam by written application to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason of missing the exam have to be presented with the application.

- For **final exam**, there will be NO makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the final exam, the student MUST get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor.**
- All mobile phones MUST be turned to silent mode during class and exam period.
- There is **zero tolerance for cheating** in exam. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university.**